Bartlett adjustment for random effects meta-analysis of event outcomes with small samples

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Abstract. Stijnen et al. (2010) describes 4 random effects models for meta-analyses of event outcomes. Inference for these models is based on an asymptotic chi-squared approximation. We describe a procedure for Bartlett correction of the confidence interval, improving the convergence rate from $1/n$ to $1/n^2$ under ideal conditions. We apply the proposed procedure to meta-analyses considered by Stijnen et al. (2010) on anti-infection devices, finding a differences in CI lengths up to 10%.

Keywords: Meta-analysis, Bartlett correction, Generalized linear mixed models

1 Introduction

There is currently underway a move to favor generalized linear mixed models in carrying out meta-analyses of event outcomes, particularly when sample sizes are small or events are rare. GLMMs avoid many of the dubious assumptions and devices, such as perfect knowledge of the study standard deviations and zero-cell corrections, required by the traditional alternative, inverse-variance weighting. However, inferences under GLMMs are still based on an asymptotic chi-squared approximation. Therefore, as in the case of inverse-variance weighted estimation, the conclusions of smaller meta-analyses are suspect, though smaller meta-analyses are where the use of GLMMs is often recommended. A classical remedy is Bartlett correction, which leverages modelling assumptions to improve the efficiency of the testing procedure. When the modelling assumptions are met, the improvement is typically a jump in the rate of asymptotic convergence from $1/n$ to $1/n^2$.

While Bartlett corrections have long been applied to generalized linear models (Cordeiro, 1983), there do not appear to be extensions to even the simple GLMMs that are used in meta-analysis.[[verify this]] The reason for the reluctance appears to be computational difficulty (Hanada and Sugimoto, 2026). While computer algebra systems have long been applied to compute the third- and fourth-order derivative arrays needed for Bartlett correction of GLMs (Cordeiro, 1993; McCullagh, 1987; Andrews and Stafford, 2000), closed-form expressions for GLMM Bartlett corrections are unavailable for any of the commonly used models other than the normal-normal (Noma, 2011; Michael, 2026). A combination of automatic symbolic manipulation and numerical approximation appears necessary. We argue that with

proper design these difficulties can be overcome and consumer-grade hardware is sufficient for practical implementation.

In the next section we present four event data meta-analysis models, following Stijnen et al. (2010)'s exposition. We then summarize the method of Bartlett correction, and discuss some of the underlying assumptions. In Section 3 we offer a practical approach to estimation of the Bartlett correction applicable to the four models described in Section 2. In Section 4 we examine by simulation the effectiveness and robustness of the corrected test, finding that the proposed approximations are suitable for achieving the desired rate boost and identifying common model violations that may neutralize the advantage. In Section 5 we re-visit the data analyses of Stijnen et al. (2010). We find that, though the advantage of Bartlett correction may be negligible for moderate or large meta-analyses, it may be clinically relevant in smaller meta-analyses.

2 Background

2.1 Statistical models

We consider the random effects model for a meta-analysis consisting of n studies. In this model, a Gaussian random variable models the effects $\theta_1, \ldots, \theta_n$, of the individual studies in the meta-analysis,

$$\theta_i = \mu_0 + \tau_0 Z_i, \qquad Z_i \sim N(0,1)$$

The target of inference in the meta-analysis is μ_0, with τ_0^2 a nuisance parameter. The distribution of the data given the random effect depends on the type of analysis being conducted. Following Stijnen et al. (2010) we consider four types of analyses, determined by whether the studies included are single-arm or two-arm trials, and whether the estimand is based on proportions or incidence.

1. Single arm, proportions. Each study reports the number of events Y_i and total number of trials m_i. The statistical model is binomial

$$\begin{aligned} f_{m_i}(y_i \mid Z_i) &= \binom{m_i}{y_i} p^{y_i}(1-p)^{m_i - y_i} \\ p = p(\mu_0, \tau_0^2) &= \operatorname{expit}(\mu_0 + \tau_0 Z_i) \end{aligned} \tag{1}$$

   The target of inference μ_0 represents the log-odds of an event. The number of trials m_i is treated as a parameter of the conditional PMF f_{m_i}, which therefore typically varies across the studies in the meta-analysis.

2. Two arms, proportions. Each study reports, for treatment and control arm, the number of events $Y_i^{(T)}$ and $Y_i^{(C)}$ and the total number of trials $m_i^{(T)}$ and $m_i^{(C)}$. The statistical model is conditional on the total number of events $S_i = Y_i^{(T)} + Y_i^{(C)}$, under which $Y_i^{(T)}$

follows Fisher's non-central hypergeometric distribution

$$f_{m_i^{(T)},m_i^{(C)},S_i}(y_i^{(T)} \mid Z_i) = \frac{\binom{m_i^{(T)}}{y_i^{(T)}}\binom{m_i^{(C)}}{S_i - y_i^{(T)}}\omega^{y_i^{(T)}}}{\sum_{\max(0,S_i-m_i^{(C)})}^{\min(S_i,m_i^{(T)})}\binom{m_i^{(T)}}{y}\binom{m_i^{(C)}}{S_i-y}\omega^{y}} \tag{2}$$
$$\omega = \omega(\mu_0, \tau_0^2) = \exp(\mu_0 + \tau_0 Z_i)$$

The target of inference μ_0 represents the ratio of the treatment to control log-odds. The parameters of the conditional PMF are $m_i^{(T)}, m_i^{(C)}$, and S_i.

3. Single arm, incidence rate. Each study reports the number of events Y_i and the duration of observation T_i. The model is Poisson,

$$f_{T_i}(y_i \mid Z_i) = \exp(-\lambda)\frac{\lambda^{y_i}}{y_i!} \tag{3}$$
$$\lambda = \lambda(\mu_0, \tau_0^2) = T_i \exp(\mu_0 + \tau_0 Z_i)$$

The target of inference μ_0 represents the log of the incidence rate. The follow-up time T_i is treated as a parameter of the conditional PMF.

4. Two arms, incidence rates. Each study reports, for treatment and control arm, the number of events $Y_i^{(T)}$ and $Y_i^{(C)}$ and the duration of observation $T_i^{(T)}$ and $T_i^{(C)}$. Conditionally on $S_i = Y_i^{(T)} + Y_i^{(C)}$, $Y_i^{(T)}$ is binomial

$$f_{T_i^{(T)},T_i^{(C)},S_i}(y_i^{(T)} \mid Z_i) = \binom{S_i}{y_i^{(T)}} p^{y_i^{(T)}}(1-p)^{S_i - y_i^{(T)}} \tag{4}$$
$$p = p(\mu_0, \tau_0^2) = \text{expit}(\log(T_i^{(T)}/T_i^{(C)}) + \mu_0 + \tau_0 Z_i)$$

The target of inference is the logarithm of the ratio of the incidence rates between the two arms. The sum of events S_i and the periods $T_i^{(T)}$ and $T_i^{(C)}$ are teated as parameters of the conditional PMF.

In all cases the likelihood for study i has the form

$$L_i(\mu, \tau^2 | Y_i) = \int_{-\infty}^{\infty} f_i(Y_i|z)\phi(z)dz, \qquad i = 1, \ldots n \tag{5}$$

where $f_i(y|z)$ is study i's PMF conditional on a Gaussian random effect, given above. Furthermore, the model parameters (μ, τ^2) only enter the likelihood (5) through a parameter of the conditional PMF, the mean λ in the case of the Poisson, the success probability p in the cases of the binomial, or the odds ratio ω in the case of the non-central hypergeometric.

The likelihood ratio test of the null $\mu = \mu_0$ is based on the test statistic

$$W = 2(l(\hat{\mu}, \hat{\tau}^2) - l(\mu_0, \hat{\tau}_0^2))$$

where

$$l(\mu, \tau^2) = \sum_{i=1}^{n} \log L_i(\mu, \tau^2 | Y_i) \tag{6}$$

is the total log-likelihood, $(\hat{\mu}, \hat{\tau}^2)$ maximizes $(\mu, \tau^2) \mapsto \sum_{i=1}^n \log L_i(\mu, \tau^2|Y_i)$, and $\hat{\tau}_0^2 = \hat{\tau}_0^2(\mu)$ maximizes $\tau^2 \mapsto \sum_{i=1}^n \log L_i(\mu_0, \tau^2|Y_i)$. In routine applications W converges to a chi-squared distribution under the null hypothesis, by Wilks' Theorem. Relying on this asymptotic approximation a confidence interval for μ_0 can be formed by inverting the test statistic.

2.2 Bartlett correction

Under a range of conditions citations the mean of a correctly specified LR statistic ((ref)) admits an expansion in integral powers of $1/n$

$$\mathbb{E}(W) = 1 + b/n + O(1/n^2).$$

Therefore, the Bartlett-corrected statistic

$$W_B = \frac{W}{1 + b/n} \tag{7}$$

has a mean agreeing with that of a chi-squared random variable up to order $O(1/n^2)$. The work of Bartlett (1937); Lawley (1956) shows further that, under appropriate conditions, all other moments, and so also the entire distribution, also agree with a chi-squared up to order $O(1/n^2)$. This rate improvement holds even when MLEs are plugged in for any unknown parameters appearing in the bias term b of the expansion (Barndorff-Nielsen and Hall, 1988). As with the LR statistic, the adjusted statistic may be inverted to obtain CIs and p-values.

Lawley (1956) provides a formula for the correction b appearing in (7) as sums and products of the components of the inverse Fisher information for (6) and certain tensors $k_r^{(s)}$. Being a small set of sums and products this formula does not introduce numerical or computational difficulties that aren't already present in the basic tensor components input into the formula, so we focus here on the latter. To this end, let D_r denote partial differentiation with respect to a sequence of the parameters μ and τ^2 given by r. For example, $D_{\mu,\tau^2} \sum_{i=1}^n \log L_i(\mu, \tau^2 \mid Y_i)$ denotes the off-diagonal term of the observed information matrix for (6). Define $k_r = k_r^{(0)} = \mathbb{E} D_r \sum_{i=1}^n \log L_i(\mu, \tau^2 \mid Y_i)$ and $k_r^{(s)} = D_s k_r$. In this notation, for example, the Fisher information matrix has components $k_{\mu,\mu}, k_{\mu,\tau^2}, k_{\tau^2,\mu}, k_{\tau^2,\tau^2}$. Lawley's formula for the Bartlett correction requires all terms k_r where r is of length 2, 3, or 4, and all terms $k_r^{(s)}$ where r is of length 3 and s of length 1 or r is of length 2 and s of length 1 or 2. Taking into account permutation symmetry of the order of differentiation, there are 35 such terms in a model with one parameter of interest and one nuisance paramter.

By linearity of expectation and differentiation, it suffices to compute the terms $k_r^{(s)}$ for the unit log-likelihoods $\log L_i(\mu, \tau^2 \mid Y_i), i = 1, \ldots, n$. Let us consider a single study from the meta-analysis, dropping the subscript i from the conditional PMFs and likelihoods. Then the general tensor component needed for the computation of Bartlett's correction is

$$k_r^{(s)} = \sum_y D_s \left(\left(D_r \log \int f(y|z)\phi(z)dz \right) \int w_k f(y|z)\phi(z)dz \right) \tag{8}$$

The outer sum is taken over the support of Y.

2.3 Assumptions

Among the assumptions underlying the above development we focus on three prominent ones. The first two stem from assumptions of the basic LR tests described in ((stijnen)), while the third is specific to Bartlett correction.

1. The Bartlett correction requires correct specification of the likelihood, in particular, gaussianity of the unobserved random effect. [[we refer to sitjnen to other modeling assumptions implicit in ((ref 1–4 above)), such as ... .]] While the LR test itself depends on gaussianity for valid inference ((citation showing effects of bad estimatio of tau2)), the Bartlett correction may be less robust to deviations.

2. Wilks' theorem, underlying the chi-squared approximation to the LR statistic ((ref)), applies to points in the interior of the parameter space, excluding the fixed effects model $\tau_0^2 = 0$. It is possible the coverage rate at a given meta-analysis sample size n looks unaffected by low heterogeneity, as has been suggested (Jackson et al., 2018; Bakbergenuly and Kulinskaya, 2018)verify, while the more delicate rate analysis carried out here betrays an impact. After all, the LR test for τ^2 is known to be inconsistent at the boundary $\tau^2 = 0$ (Self and Liang, 1987) while the parameters (μ, τ^2) are dependent non-orthogonal in the models (1),(2),(3),(4).

3. The arguments under which the Bartlett correction delivers a $1/n^2$ rate are usually justified by Cramér's condition for the underlying random variables[[cite janson]]. This condition excludes lattice variables, such as are considered here. Consequently, while Bartlett corrections have been derived for statistics based on lattice random variables since the 1980s (Cordeiro, 1983), their effectiveness does not rest on a firm theoretical foundation and in fact does not appear to be understood even in basic cases Frydenberg and Jensen (1989); Jensen (1993); Götze and van Zwet (2006).

We have no theoretical contibutions to make here and will examine model robustness and the effect of discreteness by simulation.

3 Computing the tensor components $k_r^{(s)}$

3.1 Overview

To compute the tensor component (8), we approximate the integrals by a quadrature rule. Since all the likelihoods (1)–(4) involve integrating against a Gaussian, Gauss-Hermite quadrature is a natural choice. This and similar methods (Liu and Pierce, 1994) provide weights $w_1, \ldots, w_{n_{GH}}$ and nodes $z_1, \ldots, z_{n_{GH}}$ in the domain of integration allowing the Gaussian integral of smooth integrands such as appear in ((ref)) to be approximated by a weighted sum,

$$k_r^{(s)} \approx \sum_y D_s \left(\left(D_r \log \sum_j w_j f(y|z_j) \right) \sum_k w_k f(y|z_k) \right) \tag{9}$$

The weights and nodes of the quadrature may in general depend on the unknown parameters μ and τ^2. To simplify the above derivatives we assume they are constant with respect to the parameters, as discussed further below. The number of nodes n_{GH} controls the quality of the approximation and introduces a tuning parameter into the proposed method. As in (8), the outer sum is taken over the support of Y, which is a finite set in all the models considered here other than the Poisson-normal model ((ref)). The Poisson model requires a truncation point for computation, which introduces a further tuning parameter.

Carrying out the differentiations in (9) yields rational expressions in derivatives of the weighted sums $\sum_j w_j f(y|z_j)$, say, $F_{kl}(y) = \sum_j w_j \frac{d^k}{d\mu^k} \frac{d^l}{d(\tau^2)^l} f(y|z_j)$. These 35 expressions for $k_r^{(s)}$ in terms of the F_{kl} are the same for any model based on a likelihood of the form ((ref)) and may be pre-computed. For example, $k_{\mu,\mu} = \sum_y (F_{20}(y) - (F_{10}(y)^2/F_{00}(y)))$, and $k_{\mu,\tau^2}^{(\tau^2)} = \sum_y (F_{01}(y)^2 F_{10}(y)/F_{00}(y)^2 - (F_{02}(y)F_{10}(y) + F_{01}(y)F_{11}(y))/F_{00}(y) + F_{12}(y))$.

The derivatives $F_{jk}(y)$ of the model-specific conditional PMFs that appear in the 35 tensor components $k_r^{(s)}$ number 15: $F_{jk}(y)$ for $0 \leq j + k \leq 4$. The form of these derivatives may be derived independently of data but their specific values in general depend on parameters of the conditional PMF ((see above)), and must typically be computed for each study of a meta-analysis. However, since the conditional PMFs only depend on the model parameters (μ, τ^2) through the PMF's distributional parameters, the derivatives $F_{jk}(y), 0 \leq j + k \leq 4$, may be re-written as linear combinations [[linear for binomial and hyperg also??]] of $F_{00}(y) = \sum_j w_j f(y|\theta_j)$ using standard recurrences (Johnson et al., 2005). The advantage of this form is that the terms $F_{00}(y)$ involve only the PMF itself, for which optimized and numerically stable routines are available. For example, two applications of the Poisson PMF recurrence $\frac{d}{d\lambda} f_\lambda(y \mid Z) = f_\lambda(y-1 \mid Z) - f_\lambda(y \mid Z), y \geq 1$, yield, for model ((ref poisson/incidence rate model)),

$$\begin{aligned} F_{11}(y) &= \sum_j w_j \frac{d^2}{d\mu d\tau^2} f_{\lambda(\mu,\tau^2)}(y \mid z_j) \\ &= \frac{d\lambda}{d\mu}\frac{d\lambda}{d\tau^2} F_{00}(y-2) + (\frac{d^2\lambda}{d\mu d\tau^2} - 2\frac{d\lambda}{d\mu}\frac{d\lambda}{d\tau^2}) F_{00}(y-1) + \frac{d\lambda}{d\mu}\frac{d\lambda}{d\tau^2} F_{00}(y), \qquad y \geq 2 \end{aligned}$$

The approach in brief is, for each of the n studies in the meta-analysis, 1. evaluate the 15 vectors $F_{jk}(y), y \in \text{supp}(Y)$, with $\text{supp}(Y)$ truncated in the case of Poisson Y, using only off-the-shelf routines for the conditional PMF (Poisson, binomial, or hypergeometric) and basic arithmetic, 2. compute the 35 $k_r^{(s)}$ terms using only the computed 15 $F_{jk}(y)$ vectors and basic arithmetic. The 15 $F_{jk}(y)$ terms can be computed in parallel, and once they are computed the 35 $k_r^{(s)}$ terms can be computed in parallel. As the studies are independent the entire computation of the study-level Bartlett correction can be parallelized across the n studies, as well.

3.2 Tuning parameters

We discuss the main choices made in applying the proposed method to a give dataset: the number of nodes to be used in the quadrature rule, whether the nodes should be chosen

using a fixed or adaptive rule, and, in the case of the Poisson-normal model, the truncation point for the support.

1. The main tuning parameter of the proposed method is n_{GH}, the number of nodes in the quadrature aproximation. We propose tuning this parameter heuristically, by examining the stability of the CI under a range of choices. Typical choices used in the datasets below, leading to apparent convergence up to .001 absolute error in the endpoints, are 100–200 nodes for fixed rule quadrature and 10–30 for adaptive rule quadrature.[[ref fig if available]]

2. Two broad variants fo GH quadrature are fixed and adaptive. Fixed GH uses the same choice of nodes centered at 0, irrespective of the integrand. On the other hand, adaptive GH transforms the nodes to take into account the mode and scale of the integrand. The drawback of adaptive quadrature is the cost of repeatedly numerically estimation of the mode and scale of the integrand of (5). For example, computing the MLEs of under models (1)–(4) by Newton-Raphson requires evaluating the integrals in (6) at a sequence of values (μ, τ^2), and under adaptive quadrature each such evaluation requires estimating the mode and scale of a new integrand, leading to nested optimizations. This cost is mitigated by the fewer nodes generally required by adaptive quadrature ((ref table or fig for bn-prop)).

   A further difficulty with adaptive quadrature is conceptual. Since the adaptive nodes and weights depend on the parametes (μ, τ^2) through the mode and scale, they are subject to the derivatives in (9), which is incompatible with the assumptions of the proposed method. Ignoring this dependence involves a tacit assumption that the mode and scale of the derivatives of the integrand are similar to the integrand's. However, this simplified approach leads to efficient implementation of adaptive GH. The only type of integrand for which a mode must then be computed is of the form (5). As all the conditional PMFs belong to the exponential family, these integrands are strictly log-concave and unimodal, and the mode and scale may be obtained quickly using Newton-Raphson.

   In most of the data analysis examples in Section 5, there is little difference in performance between CIs returned by fixed and adaptive quadrature. In general one expects adaptive quadrature to be valuable when the integrand charges the support of Y far from 0, which is where fixed rule nodes are centered. Consistent with this heuristic, the Poisson-normal model benefits heavily from adaptive quadrature ((ref picture or table)).

3. Since the Poisson-normal model has infinite support, implementation of the proposed method requires a truncation point. Once a GH rule has been chosen, one can choose the truncation point large enough that the weighted tails are as small as desired. However, it may be difficult to know in general what is to be deemed negligible for a given data set, and moreover the truncation point often influences the choice of n_{GH}. maybe given example of higher oscillation obsreved in pn data analysis For the Poisson-normal model we therefore suggest examining a matrix of CIs varying both parameters to identify the limiting CI. See Section 5 for an example.

4 Simulation

We examine by simulation the rate at which the LR and Bartlett-corrected LR statistics converge to the reference chi-squared distribution under the four models (1)–(4). The purpose is three-fold: 1. to investigate whether the quadrature approximations used by the proposed method are sufficient, 2. to investigate whether the Bartlett correction is effective as applied to this type of lattice data, as there are few theoretical results here, and 3. to assess the robustness of any improvement in the rate offered by Bartlett correction to deviations from the assumed non-degenerate Gaussian model for the random effect. Results are given in Table 1 and Figure 1.

4.1 Design

To make the simulation results relevant, the simulated meta-analyses are constructed to resemble the meta-analysis datasets of Tables 4 and 5 analyzed by Stijnen et al. (2010) and re-analyzed in Section 5. The true effect μ_0 and heterogeneity τ_0^2 are taken to be the MLEs for the data obtained by Stijnen et al. (2010), and the distributional parameters are obtained by sampling with replacement from the Tables 4 and 5. For example, for the control-arm binomial-normal model (1), a simulated meta-analysis of size n is drawn by, first, sampling with replacement n trial sizes $m^{(C)}$ from the 18 control arm study sizes given by the corresponding column of Table 4; and second, sampling n effects Y under the binomial likelihood (1) using the sampled trial sizes and the parameters $(\mu_0, \tau_0^2) = (-3.50, .81)$ taken from the MLE calculations of Stijnen et al. (2010). Meta-analyses of sizes $5, \ldots, 15$, are sampled in this manner, and the rate of convergence of the empirical Kolmogorov-Smirnov distance to a chi-squared distribution is estimated using n^4 monte carlo repetitions at each sample size n. See Michael (2026) for details on the rate estimation.

Robustness of the gaussianity assumption is examined by perturbing a normal to a skewed distribution (Jones and Pewsey, 2009). The skew was chosen to test robustness since it drives the first-order error in CLT-type approximations. Robustness against non-degeneracy of the random effect, $\tau_0^2 > 0$, is studied in particular cases below.

In all simulations, we have used the true parameters (μ_0, τ_0^2) to calculate the Bartlett correction $b = b(\mu_0, \tau_0^2)$ perturbed by a Gaussian with SD $1/\sqrt{n}$??. In practice the MLE would be used. The practical reason is to avoid the prohibitive computational cost of re-calculating the Bartlett calculation for each of the n^4 monte carlo replicates. The justification is the result of Barndorff-Nielsen and Hall (1988) and others that the $O(1/n^2)$ Bartlett rate survives substitution of $\sqrt{n}-$consistent estimators into the Bartlett correction $b(\mu, \tau^2)$.

4.2 Results

Table 1 shows that when the model assumptions are met, the standard LR rate of $1/n$ and the Bartlett-corrected rate of $1/n^2$ appear to be achieved by three of the four models (1)–(4). As skew increases, the rate deterioriates for both statistics. [[generally lr has a slower rate than the bartlett throughout, but for the largest skew tested and the bn-prop model, the bartlett corrected statistic shows a worse rate than the uncorrected.]] We therefore conclude

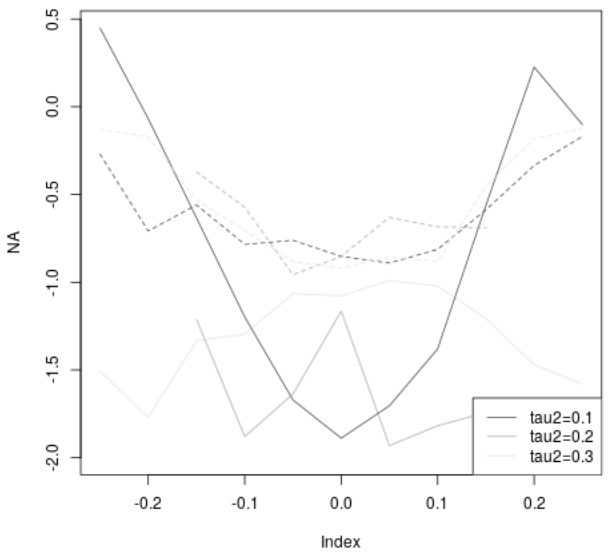


Figure 1: The effect of varying the heterogeneity parameter τ^2. The data is simulated based on ((nw2009)) and the binomial-normal model for incidence rate ratios.

that, at least in these three models, the quadrature approximations may be sufficient and that skew may affect the convergence of both corrected and uncorrected LR statistics.

The binomial-normal model for incidence rate ratios model (4) is the exception, showing a rate closer to $1/n$ than $1/n^2$ under Bartlett correction even in the idealized skew-free setting. This failure appears to be due to low heterogeneity. Rather than the MLE $\hat{\tau}^2 = 0.096$ of the original data set (Niel-Weise et al., 2008), we examine the effect of using ((tau2= ...)), maintaining other aspects of the simulation design. The results, presented in Fig. 1, show the $1/n^2$ rate being achieved as τ^2 grows to ... . ((same for 1/n rate of LR stat?)) we therefore conclude that the advantage of Bartlett correction can be impaired by low heterogeneity.

Study	**Statistic**	0	0.05	0.1	0.15	0.2
bn.prop	LR (standard)	-1.14 (0.12)	-1.14 (0.08)	-0.99 (0.08)	-0.64 (0.18)	-0.34 (0.14)
	LR (Bartlett)	-1.87 (0.35)	-1.25 (0.31)	-1.13 (0.35)	-0.29 (0.26)	0.00 (0.26)
pn	LR (standard)	-1.16 (0.19)	-1.21 (0.17)	-0.69 (0.32)	-0.59 (0.03)	-0.52 (0.21)
	LR (Bartlett)	-1.89 (0.44)	-1.87 (0.34)	-0.96 (0.36)	-0.41 (0.02)	-0.29 (0.45)
bn.irr	LR (standard)	-0.83 (0.20)	-0.91 (0.20)	-0.79 (0.19)	-0.72 (0.24)	-0.38 (0.17)
	LR (Bartlett)	-1.01 (0.21)	-1.10 (0.17)	-1.13 (0.24)	-1.15 (0.23)	-1.20 (0.36)

Table 1: Simulation results. The cells contain the medians and SDs of the log of the rate of the empirical Kolmogorov-Smirnov distance to a chi-squared distribution.

5 Data analysis

We revisit the example data analyses of Stijnen et al. (2010). Meta-analyses are carried out under the models (1)–(4) and 95% confidence intervals for the underlying effect μ_0 are calculated using the corrected and uncorrected profile LR. The results are presented in Table 2.

5.1 Proportions

Niel-Weise et al. (2007) carries out a meta-analysis on the efficacy of an anti-infection treatment for catheters. Each of the 18 included studies reports, for treatment and control arms, the total number of patients in the study and number of those who developed a catheter-related infection. See Table 4., incorporating the correction to Stijnen et al. (2010) given by ((meta-analysis guy)).

We apply the binomial-normal model for proportions (1) to obtain CIs for the log odds of infection in each arm separately and the hypergeometric-normal model (2) to obtain a CI for the log-odds ratio across the arms. In all three cases, the CI using the Bartlett-corrected profile likelihood is only slightly wider than the uncorrected, with the endpoints differing only in the hundredth's place or less. This agreement may reflect the moderate sample size of the the meta-analysis at 17–18 studies.

[[give sample plot]]

5.2 Incidence

Niel-Weise et al. (2008) also presents a meta-analysis on the efficiacy of an anti-infection treatment for catheters. Each of the 9 included studies reports, for each arm, the number of catheter-related infections observed in a population as well as the total "catheter-days," the sum over the study population of the number of days each subject used a catheter.

We apply the Poisson-normal model (3) to obtain CIs for the log incidence rate in each arm, and the binomial-normal model for incidence rate ratios (4) to obtain a CI for the log-incidence-rate ratio. For this smaller meta-analysis the difference between the corrected and uncorrected CIs is more dramatic, the corrected CIs being about 10% larger. These differences include a loss of significance at the nominal level $\alpha = .05$ for the log-IR ratio study.

The two tuning parameters required by the Poisson-normal model were selected using the matrix presented in Table 3. Taking $n_{GH} = 10$ adaptive nodes and $y_{max} = 400$ support points is sufficient for a CI with endpoints within 1/1000 of the converged value, where $n_{GH} = 100$ adaptive nodes and $y_{max} = 1000$ is assumed to produce the converged value.

6 Conclusion

We have presented a method to more accurately compute confidence intervals for meta-analyses of event outcomes. we have not observed an effect of discreteness, perhaps vindicating the many bartlett corrections produced for discrete-data models without a firm

Estimand	**Meta-analysis**	**LR CI**	**Proposed CI**	**% Diff**
log odds	stijnen-1, control arm	[-4.07, -3.00]	[-4.09, -2.98]	+3.7%
	stijnen-1, treatment arm	[-5.70, -4.20]	[-5.73, -4.19]	+2.7%
log-odds ratio	stijnen-1	[-2.21, -0.71]	[-2.25, -0.68]	+5.1%
log IR	stijnen-6, control arm	[0.90, 1.90]	[0.85, 1.95]	+10.6%
	stijnen-6, treatment arm	[0.10, 1.55]	[0.03, 1.62]	+9.8%
log-IR ratio	stijnen-6	[-1.05, -0.01]	[-1.12, 0.05]	+11.3%

Table 2: Standard and Bartlett-corrected confidence intervals for the example meta-analysis datasets considered by ((stijnen)). There is little advantage in Bartlett correction for the meta-analyses [[nw87 data]], which consists of 18 studies. There is a greater difference for the meta-analyses based on [[nw88 data]], which has half as many studies. [[switch first and second columns]]

	Poisson support cutoff $y_{\max}$				
$n_{\mathbf{GH}}$	**100**	**200**	**300**	**400**	**1000**
5	[0.9526, 1.7905]	[0.9585, 1.7592]	[0.9573, 1.7459]	[0.9568, 1.7407]	[0.9565, 1.7366]
6	[0.8713, 1.9132]	[0.8704, 1.9206]	[0.8684, 1.9236]	[0.8675, 1.9250]	[0.8669, 1.9282]
7	[0.8500, 1.9692]	[0.8525, 1.9580]	[0.8509, 1.9543]	[0.8501, 1.9529]	[0.8495, 1.9533]
8	[0.8557, 1.9431]	[0.8575, 1.9418]	[0.8558, 1.9409]	[0.8550, 1.9405]	[0.8444, 1.9418]
9	[0.8512, 1.9486]	[0.8531, 1.9471]	[0.8513, 1.9460]	[0.8505, 1.9456]	[0.8499, 1.9469]
10	[0.8496, 1.9512]	[0.8518, 1.9486]	[0.8500, 1.9473]	[0.8492, 1.9468]	[0.8486, 1.9479]
100	[0.8495, 1.9503]	[0.8516, 1.9482]	[0.8499, 1.9470]	[0.8491, 1.9465]	**[0.8485, 1.9477]**

Table 3: Convergence of the Bartlett-corrected CI for the Poisson-normal mode as its two tuning parameters are varied. [[add vertical line before last column]]

theoretical basis. we have seen that improvement offered by the proposed method deteriorates as the model assumption of a non-degenerate Gaussian random effect is progressively relaxed.

Ideally application of the bartlett correction would be a "free lunch," yielding an improved rate if the random effect is in fact Gaussian and otherwise being [[neutral]], other than perhaps computational cost. We have seen this is not quite the case. When the skew of the random effect is sufficiently far from 0, the rate of the bartlett corrected sttaistic can be worse than the uncorrected statistic. Presumably a random effect sufficiently far from a Gaussian in other respects can also lead to this reversal.

Software implementing the proposed method and the routines used to carry out the simulations are available at the author's website.

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Study	$Y^{(C)}$	$n^{(C)}$	$Y^{(T)}$	$n^{(T)}$
1	3	117	0	116
2	3	35	1	44
3	9	195	2	208
4	7	136	0	130
5	6	157	5	151
6	4	139	1	98
7	3	177	1	174
8	2	39	1	74
9	19	103	1	97
10	2	122	1	113
11	7	64	0	66
12	1	58	0	70
13	5	175	3	188
14	11	180	6	187
15	0	105	0	118
16	1	262	0	252
17	3	362	1	345
18	1	69	4	64

Table 4: Data from Niel-Weise et al. (2007) analyzed in Section 5 under the binomial-normal model for proportions and the hypergeometric-normal model. Study 15 does not contribute to the hypergeometric-normal likelihood (2), reducing the sample size to 17 for that analysis in Section 5.

Study	$Y^{(C)}$	$T^{(C)}$	$Y^{(T)}$	$T^{(T)}$
1	11	1988	7	1344
2	8	1460	8	1599
3	14	10962	3	12012
4	10	1502	6	1536
5	1	482	1	370
6	8	913	1	729
7	15	6840	17	6760
8	7	1015	3	1107
9	3	440	2	320

Table 5: Data from Niel-Weise et al. (2008) analyzed in Section 5 under the binomial-normal model for incidence rate ratios and the Poisson-normal model.